

A MoMo credit top-up completed by MTN's callback

A throwaway viewer bought 1,000 in credit with MoMo through the Nightgals API. The app sent MTN a callback address, MTN reported the result to it, and the app completed the purchase. Below is what the app logged as it sent each request to MTN, what MTN answered, and the callback MTN sent. Keys and tokens are masked by the app itself.

Purchase e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f	Bought Credit top-up, 1,000 XAF	Payer number 237689686224
Final status ● COMPLETED	MTN transaction ID 281824661	Pay → completed 3.3 s, by callback

What happened, in order

Times are UTC. The offset below each time counts from the buyer pressing "pay".

■ Buyer's app → Nightgals API ■ Nightgals → MTN ■ MTN → Nightgals ■ Nightgals updates the purchase

08:47:35.768 ■
+0.0 s

BUYER → NIGHTGALS Buyer pays for 1,000 credit with MoMo

POST /api/v1/billing/credit/top-up. The app creates a PENDING purchase and uses its ID for everything that follows.

```
{
  "amountMinor": 1000,
  "method": "MOMO",
  "payerMsisdn": "237689686224"
}
```

08:47:35.924 ■
+0.2 s

NIGHTGALS → MTN Gets an access token

MoMo → POST https://sandbox.momodeveloper.mtn.com/collection/token/ headers=[Authorization:"Basic ****", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>

08:47:36.873 ■
+1.1 s

MTN → NIGHTGALS Token issued, valid 1 hour 948 ms

MoMo ← 200 /collection/token/ 948ms body={
"access_token": "****", "token_type": "access_token", "expires_in": 3600}

08:47:36.914 ■
+1.1 s

NIGHTGALS → MTN Sends the collection request with a callback address

X-Callback-Url: <https://unneighborly-keturah-noncontingent.ngrok-free.dev/api/v1/webhooks/momo> tells MTN where to report the result. X-Reference-Id and externalId are both the purchase ID, so a retried checkout can't send a second prompt.

```
MoMo → POST https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay
headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", X-Reference-Id:"e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f", X-Callback-Url:"https://unneighborly-keturah-noncontingent.ngrok-free.dev/api/v1/webhooks/momo", Content-Type:"application/json", Content-Length:"221"] body={"payer":{"partyId":"237689686224","partyIdType":"MSISDN"},"externalId":"e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f","amount":"1000","currency":"EUR","payeeNote":"Nightgals credit topup","payerMessage":"Nightgals credit topup"}
```

08:47:37.263 ■
+1.5 s

MTN → NIGHTGALS Accepted: prompt sent to the handset 349 ms

202 means accepted, not paid. There is no body.

```
MoMo ← 202 /collection/v1_0/requesttopay 349ms body=<empty>
```

08:47:37.358 ■
+1.6 s

NIGHTGALS Purchase left PENDING, buyer told to check their phone

● **PENDING** 1.6 s

MoMo prompt sent for purchase e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f to ****6224

```
{
  "purchase": {
    "id": "e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f",
    "type": "CREDIT_TOPUP",
    "amountMinor": 1000,
    "priceDisplay": "1000",
    "currency": "XAF",
    "status": "PENDING",
    "provider": "MOMO",
    "createdAt": "2026-09-14T08:47:35.877722450Z"
  },
  "action": "PROMPT_ON_PHONE",
  "instructions": "Check your phone and approve the Mobile Money request."
}
```

08:47:38
+2.2 s

MTN → NIGHTGALS MTN calls the app back • SUCCESSFUL

POST /api/v1/webhooks/momo arrived through the tunnel within 2 s of MTN accepting the request, and the app answered 200. ngrok records this time only to the second.

```
{
  "financialTransactionId": "281824661",
  "externalId": "e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f",
  "amount": "1000",
  "currency": "EUR",
  "payer": {
    "partyIdType": "MSISDN",
    "partyId": "237689686224"
  },
  "payeeNote": "Nightgals credit topup",
  "status": "SUCCESSFUL"
}
```

08:47:38.715
+3.0 s

NIGHTGALS → MTN App confirms the status with MTN

The callback is unauthenticated, so the app never completes a purchase from it directly. It asks MTN over an authenticated call first.

```
MoMo → GET
https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay/e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"]
body=<empty>
```

08:47:39.027
+3.3 s

MTN → NIGHTGALS MTN confirms: approved • SUCCESSFUL 310 ms

```
{
  "financialTransactionId": "281824661",
  "externalId": "e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f",
  "amount": "1000",
  "currency": "EUR",
  "payer": {
    "partyIdType": "MSISDN",
    "partyId": "237689686224"
  },
  "payerMessage": "Nightgals credit topup",
  "payeeNote": "Nightgals credit topup",
  "status": "SUCCESSFUL"
}
```

08:47:39.074 ■
+3.3 s

NIGHTGALS Purchase completed and credit added • COMPLETED

Completed with MTN's transaction ID as the payment reference, then 1,000 added to the viewer's balance, 3.3 s after the buyer paid.

```
XAF 1,000 credited to a31049fe-cc72-496a-9283-c5a9136ef529 from top-up
e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f
Purchase e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f settled (CREDIT_TOPUP)
MoMo purchase e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f settled by callback
```

08:47:52.401 ■
+16.6 s

BUYER → NIGHTGALS Buyer's app sees COMPLETED; balance is 1,000

• COMPLETED

The buyer's app polls every 15 s, so it saw the change on its next check.

```
{
  "purchase": {
    "id": "e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f",
    "type": "CREDIT_TOPUP",
    "amountMinor": 1000,
    "priceDisplay": "1000",
    "currency": "XAF",
    "status": "COMPLETED",
    "provider": "MOMO",
    "completedAt": "2026-09-14T08:47:39.034139Z",
    "createdAt": "2026-09-14T08:47:35.877722Z"
  },
  "credit": {
    "balanceMinor": 1000,
    "currency": "XAF"
  }
}
```

The callback MTN sent, as received

Captured by ngrok before it reached the app. It carries no signature or credential, which is why the app asks MTN for the status before completing anything.

```
POST /api/v1/webhooks/momo → HTTP 200
Content-Type: application/json; charset=utf-8
Content-Length: 240
User-Agent: LWAC Http Client 1.0
Host: unneighborly-keturah-noncontingent.ngrok-free.dev:443
X-Forwarded-For: 212.88.125.17
X-Forwarded-Proto: https
```

```
{
  "financialTransactionId": "281824661",
  "externalId": "e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f",
  "amount": "1000",
  "currency": "EUR",
  "payer": {
    "partyIdType": "MSISDN",
```

```
"partyId": "237689686224"
},
"payeeNote": "Nightgals credit topup",
"status": "SUCCESSFUL"
}
```

What we found

The callback flow works end to end

MTN posted the result to `/api/v1/webhooks/momo` within 2 s of accepting the request. The app confirmed the status with MTN, completed the purchase with MTN's transaction ID as the payment reference, and added 1,000 to the viewer's balance, 3.3 s after the buyer paid. One purchase completed, one credit entry.

The app records XAF, while sandbox is charged EUR

The purchase is stored as 1,000 XAF, but the request to MTN says `"currency": "EUR"` because `MOMO_CURRENCY=EUR`. That's expected: sandbox only accepts EUR. The currency and target environment must switch to XAF and the live values at the same time.

The app's log

Every MoMo and billing line the running app wrote during the purchase, exactly as logged (timestamps are the machine's local time, UTC+3). The colored edge marks requests to MTN, MTN's answers and purchase updates.

10 lines · nightgals-app4.log, filtered to MoMo and billing

```
11:47:35.924 DEBUG [omcat-handler-2] MomoClient MoMo → POST
https://sandbox.momodeveloper.mtn.com/collection/token/ headers=[Authorization:"Basic ****", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>

11:47:36.873 DEBUG [omcat-handler-2] MomoClient MoMo ← 200 /collection/token/ 948ms body=
{"access_token":"****","token_type":"access_token","expires_in":3600}

11:47:36.914 DEBUG [omcat-handler-2] MomoClient MoMo → POST
https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", X-Reference-Id:"e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f", X-Callback-Url:"https://unneighborly-keturah-noncontingent.ngrok-free.dev/api/v1/webhooks/momo", Content-Type:"application/json", Content-Length:"221"] body={"payer":{"partyId":"237689686224","partyIdType":"MSISDN"},"externalId":"e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f","amount":"1000","currency":"EUR","payeeNote":"Nightgals credit topup","payerMessage":"Nightgals credit topup"}

11:47:37.263 DEBUG [omcat-handler-2] MomoClient MoMo ← 202 /collection/v1_0/requesttopay 349ms body=<empty>

11:47:37.264 INFO [omcat-handler-2] MomoPaymentProvider MoMo prompt sent for purchase e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f to ****6224

11:47:38.715 DEBUG [omcat-handler-4] MomoClient MoMo → GET
https://sandbox.momodeveloper.mtn.com/collection/v1_0/requesttopay/e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f headers=[Authorization:"Bearer ****", X-Target-Environment:"sandbox", Ocp-Apim-Subscription-Key:"****", Content-Length:"0"] body=<empty>
```

```
11:47:39.027 DEBUG [omcat-handler-4] MomoClient MoMo ← 200 /collection/v1_0/requesttopay/e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f 310ms body={"financialTransactionId":"281824661","externalId":"e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f","amount":"1000","currency":"EUR","payer":{"partyIdType":"MSISDN","partyId":"237689686224"},"payerMessage":"Nightgals credit topup","payeeNote":"Nightgals credit topup","status":"SUCCESSFUL"}

11:47:39.073 INFO [omcat-handler-4] CreditService XAF 1,000 credited to a31049fe-cc72-496a-9283-c5a9136ef529 from top-up e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f

11:47:39.074 INFO [omcat-handler-4] BillingService Purchase e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f settled (CREDIT_TOPUP)

11:47:39.088 INFO [omcat-handler-4] MomoCallbackController MoMo purchase e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f settled by callback
```

The buyer's API calls

What a mobile client would send. The purchase was polled every 15 seconds until it changed.

UTC	Call	HTTP	Took	Purchase status
08:47:30.112	POST /api/v1/auth/register	201	5.6s	
08:47:35.744	GET /api/v1/billing/payment-methods	200	24ms	
08:47:35.768	POST /api/v1/billing/credit/top-up	200	1.6s	● PENDING
08:47:37.358	GET /api/v1/billing/purchases/{purchaseId}	200	42ms	● PENDING
08:47:52.401	GET /api/v1/billing/purchases/{purchaseId}	200	15ms	● COMPLETED
08:47:52.416	GET /api/v1/billing/credit	200	21ms	

Stored afterwards

Read from the local Postgres after the run.

id	e8ecb2f9-197b-4ec6-a5bc-64ac389aa49f
type	CREDIT_TOPUP
status	COMPLETED
provider	MOMO
amount_minor	1000
currency	XAF

payer_msisdn	237689686224
provider_reference	281824661
created_at	2026-09-14 08:47:35.877 UTC
completed_at	2026-09-14 08:47:39.034 UTC
credit_entries	1 · TOPUP · 1000 XAF · 08:47:39.054 UTC

How this was run

- Local app, commit 7e8afce, against MTN's sandbox, started with PAYMENT_PROVIDERS=momo.
- An ngrok tunnel gave the local app a public HTTPS address, and MOMO_CALLBACK_URL was set to its /api/v1/webhooks/momo address.
- A sandbox API user registered to the tunnel's host, since MTN only calls back to the registered host.
- Buyer: a new viewer account (momo-test-...@example.com). Payer: sandbox test number 237689686224, which MTN always approves.

Run on 2026-09-14 against sandbox.momodeveloper.mtn.com from a local Nightgals build. The buyer account was created for this test only. Keys, tokens and the account password never appear here: the app masks MoMo credentials in its own log.